Perfect. Now I can explain these two notebooks in a way that you'll be able to explain them confidently in an interview.

1️⃣ 09_mlflow_logging.ipynb

What is this file?

This is the **Producer Notebook**.

Its job is:

```
Train Model
↓
Evaluate Model
↓
Store Results in MLflow
```

Think of it as a machine that creates experiment records.

Real Life Example

Suppose you're testing:

```
XGBoost Model A
XGBoost Model B
XGBoost Model C
```

Without MLflow:

```
Notebook
Excel
Screenshots
Memory
```

Everything gets messy.

With this notebook:

```
Model
Parameters
Metrics
```

```
Plots
Artifacts
```

are automatically saved.

What happens inside this notebook?

Step 1

Imports libraries

Python

```python
import mlflow
import mlflow.sklearn
```

Purpose:

```
Enable experiment tracking
```

Step 2

Configure MLflow

Python

```python
mlflow.set_tracking_uri(
    "sqlite:///mlflow.db"
)
```

Meaning:

```
Store all experiment information
inside mlflow.db
```

Think:

```
MySQL Database
```

but for ML experiments.

Step 3

Load Data

Python

```python
train_fe.csv
```

This is your final feature-engineered dataset.

Contains:

```
EXT_SOURCE_MEAN
CREDIT_TERM
AGE
...
```

Step 4

Train Final XGBoost

Python

```python
best_xgb.fit(...)
```

Purpose:

```
Build final model
```

Step 5

Generate Predictions

Python

```python
predict()

predict_proba()
```

Purpose:

```
Evaluate model performance
```

Step 6

Calculate Metrics

Python

```python
Precision

Recall

F1

ROC-AUC
```

Example:

```
Precision = 0.2388
Recall    = 0.4268
F1        = 0.3063
ROC-AUC   = 0.7533
```

Step 7

Threshold Optimization

Your notebook also includes:

```
Threshold Selection
```

Purpose:

```
Find best probability cutoff
```

instead of always using:

```
0.50
```

Step 8

Log Parameters

Example:

Python

```python
mlflow.log_param(
    "max_depth",
    5
)
```

Stored:

```
max_depth = 5
learning_rate = 0.05
n_estimators = 300
```

Question Answered:

```
How was this model trained?
```

Step 9

Log Metrics

Example:

Python

```python
mlflow.log_metric(
    "roc_auc",
    0.7533
)
```

Stored:

```
ROC-AUC
Precision
Recall
F1
```

Question Answered:

```
How good was this model?
```

Step 10

Log Artifacts

Example:

```python
confusion_matrix.png

roc_curve.png
```

Stored in MLflow.

Question Answered:

```
What visual evidence do we have?
```

Step 11

Log Model

```python
mlflow.sklearn.log_model(...)
```

Stored:

```
Final Trained XGBoost
```

Question Answered:

```
Can I reuse this model later?
```

Answer:

```
Yes
```

Final Purpose of 09_mlflow_logging.ipynb

This notebook creates:

```
Experiment
Runs
Metrics
Parameters
Artifacts
Model
```

inside MLflow.

Think:

```
Data Producer
```

2️⃣ 07_mlflow_tracking.ipynb

What is this file?

This is the **Consumer Notebook**.

Its job is:

```
Read information
already stored in MLflow
```

Think:

```
09_mlflow_logging.ipynb
        ↓
```

```
Stores Data

07_mlflow_tracking.ipynb
        ↓
Reads Data
```

What happens inside this notebook?

Step 1

Create Client

Python

```python
client = mlflow.tracking.MlflowClient()
```

Purpose:

```
Connect to MLflow
```

Think:

```
Database Connection
```

Step 2

List Experiments

Python

```python
search_experiments()
```

Purpose:

```
Show all projects
```

Example:

ID	Name
1	Home Credit Risk

Question Answered:

```
Which projects exist?
```

Step 3

Show Runs

Python

Run

```python
search_runs()
```

Purpose:

```
Show all training attempts
```

Example:

Run	ROC-AUC
Run 1	0.74
Run 2	0.75
Run 3	0.73

Question Answered:

```
How many experiments were performed?
```

Step 4

Metrics Visualization

Notebook creates graphs.

Example:

```
ROC-AUC
vs
max_depth
```

Question Answered:

```
Which parameters improve performance?
```

Step 5

Best Run Analysis

Notebook identifies:

```
Best Model
```

based on:

```
Highest ROC-AUC
```

or another metric.

Question Answered:

```
Which experiment won?
```

Final Purpose of 07_mlflow_tracking.ipynb

This notebook:

```
Reads MLflow Data
Visualizes Results
```

```
Compares Runs
Finds Best Model
```

Think:

```
Analytics Dashboard
```

Simple Analogy

Imagine you're a scientist.

09_mlflow_logging.ipynb

```
Lab Experiment
```

You:

```
Perform Experiment
Record Observations
Store Results
```

07_mlflow_tracking.ipynb

```
Research Analysis
```

You:

```
Read Results
Compare Experiments
Find Best Result
```

Interview Explanation

If an interviewer asks:

> Why did you create two MLflow notebooks?

You can answer:

> "I separated experiment logging and experiment analysis. The logging notebook trained the final XGBoost model and stored parameters, metrics, artifacts, threshold optimization results, and the trained model into MLflow. The tracking notebook then connected to MLflow, retrieved all experiment runs, visualized metrics, compared models, and identified the best-performing run. This separation mimics real-world MLOps workflows where model training and experiment monitoring are handled independently."

That's a very strong industry-style answer and shows you understand not just ML, but the beginnings of MLOps as well. 🚀